

## YOUYANG, China

WMO: 576330

Lat: 28.8219N

Lon: 108.7671E

Elev: 2716

StdP: 13.31

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 1             | 27.4       | 29.5 | 17.5                          | 14.7 | 36.6 | 20.9 | 17.3 | 35.2 | 10.4                  | 37.9 | 9.2 | 39.4 | 2.6                   | 0    | 0.320 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 13.5                   | 90.8            | 74.1 | 88.8 | 73.8 | 86.8 | 73.4 | 76.9                | 85.5 | 76.0 | 84.2 | 75.2 | 83.0 | 4.3                  | 140  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 74.7                            | 143.6 | 80.3 | 73.8 | 139.6 | 79.6 | 73.1 | 136.1 | 79.0 | 42.7          | 85.7 | 41.7 | 84.5 | 40.9 | 83.1 | 80.8              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |      |  |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|------|--|
| 1%                | 2.5% | 5%  | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |      |  |
|                   |      |     | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |      |  |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |      |  |
| 9.5               | 7.7  | 6.9 | DB                         | 25.2 | 94.5               | 3.4 | 2.0                                                | 22.7 | 96.0       | 20.7 | 97.1       | 18.8 | 98.2       | 16.3 | 99.7 |  |
|                   |      |     | WB                         | 24.1 | 78.3               | 2.6 | 1.2                                                | 22.3 | 79.2       | 20.8 | 79.9       | 19.3 | 80.6       | 17.5 | 81.6 |  |

## Monthly Climatic Design Conditions

|                                                                               |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|-------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                               |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                           | DBAvg       | 60.0   | 39.6  | 43.1  | 50.7  | 60.3  | 67.4  | 73.0  | 78.3  | 77.8  | 70.8  | 61.5  | 52.8  | 43.2 |
|                                                                               | DBStd       | 14.40  | 5.90  | 6.95  | 7.07  | 6.35  | 4.85  | 4.45  | 3.55  | 4.15  | 5.42  | 5.56  | 6.13  | 5.16 |
|                                                                               | HDD50       | 874    | 325   | 211   | 77    | 5     | 0     | 0     | 0     | 0     | 0     | 1     | 39    | 216  |
|                                                                               | HDD65       | 3237   | 787   | 614   | 445   | 167   | 31    | 2     | 0     | 0     | 12    | 136   | 367   | 676  |
|                                                                               | CDD50       | 4513   | 3     | 17    | 100   | 313   | 540   | 691   | 877   | 861   | 625   | 358   | 123   | 5    |
|                                                                               | CDD65       | 1399   | 0     | 0     | 2     | 25    | 106   | 243   | 412   | 396   | 186   | 28    | 1     | 0    |
|                                                                               | CDH74       | 9972   | 0     | 2     | 22    | 173   | 617   | 1438  | 3347  | 3161  | 1079  | 129   | 4     | 0    |
|                                                                               | CDH80       | 3218   | 0     | 0     | 3     | 29    | 137   | 392   | 1172  | 1182  | 288   | 15    | 0     | 0    |
| Wind                                                                          | WSAvg       | 2.5    | 2.4   | 2.5   | 2.6   | 2.5   | 2.5   | 2.4   | 2.5   | 2.6   | 2.5   | 2.3   | 2.3   | 2.3  |
| Precipitation                                                                 | PrecAvg     | 52.4   | 1.2   | 1.5   | 2.5   | 4.8   | 8.0   | 9.3   | 7.5   | 5.9   | 3.9   | 4.4   | 2.6   | 0.9  |
|                                                                               | PrecMax     | 68.2   | 2.2   | 2.9   | 4.4   | 8.1   | 12.9  | 22.1  | 15.6  | 11.3  | 9.8   | 8.0   | 6.3   | 2.2  |
|                                                                               | PrecMin     | 37.7   | 0.3   | 0.1   | 1.0   | 1.1   | 3.9   | 2.3   | 1.5   | 1.1   | 0.4   | 1.9   | 0.2   | 0.1  |
|                                                                               | PrecStd     | 8.3    | 0.6   | 0.9   | 0.9   | 1.7   | 2.5   | 4.4   | 3.7   | 2.7   | 1.9   | 1.9   | 1.3   | 0.6  |
| Monthly Design<br>Dry Bulb and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 61.2  | 67.5  | 76.9  | 83.7  | 87.2  | 89.8  | 93.3  | 93.6  | 90.2  | 82.1  | 73.6  | 62.5 |
|                                                                               |             | MCWB   | 49.7  | 55.1  | 61.4  | 65.7  | 67.2  | 71.3  | 74.8  | 74.1  | 72.6  | 65.8  | 59.5  | 49.9 |
|                                                                               | 2%          | DB     | 55.2  | 60.5  | 70.3  | 79.1  | 83.8  | 87.1  | 90.7  | 91.1  | 86.3  | 77.6  | 69.4  | 58.0 |
|                                                                               |             | MCWB   | 46.7  | 50.9  | 58.2  | 64.3  | 66.9  | 72.4  | 74.9  | 74.2  | 72.0  | 63.8  | 57.9  | 47.7 |
|                                                                               | 5%          | DB     | 51.1  | 56.1  | 65.7  | 75.0  | 80.7  | 84.7  | 88.6  | 88.9  | 83.4  | 73.7  | 65.4  | 54.1 |
|                                                                               |             | MCWB   | 44.2  | 49.4  | 55.7  | 62.9  | 66.9  | 72.3  | 74.7  | 74.1  | 70.6  | 63.1  | 56.4  | 45.8 |
|                                                                               | 10%         | DB     | 48.2  | 52.9  | 61.4  | 70.8  | 77.3  | 81.8  | 86.4  | 86.8  | 80.2  | 70.3  | 61.8  | 50.9 |
|                                                                               |             | MCWB   | 42.6  | 47.9  | 53.9  | 61.9  | 66.5  | 71.6  | 74.5  | 73.8  | 69.2  | 62.1  | 55.6  | 45.1 |
| Monthly Design<br>Wet Bulb and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 52.8  | 57.3  | 65.8  | 70.0  | 73.8  | 76.6  | 78.3  | 78.0  | 75.5  | 70.9  | 63.8  | 52.7 |
|                                                                               |             | MCDB   | 58.2  | 65.3  | 72.8  | 78.7  | 80.4  | 84.0  | 87.0  | 88.2  | 84.7  | 76.6  | 69.6  | 58.7 |
|                                                                               | 2%          | WB     | 48.1  | 53.1  | 60.0  | 67.3  | 71.8  | 75.4  | 77.2  | 76.6  | 74.1  | 67.2  | 60.7  | 49.7 |
|                                                                               |             | MCDB   | 53.3  | 57.6  | 67.1  | 74.4  | 78.0  | 82.2  | 85.8  | 85.7  | 82.4  | 72.4  | 65.0  | 54.7 |
|                                                                               | 5%          | WB     | 45.6  | 50.9  | 57.2  | 64.9  | 69.8  | 74.5  | 76.3  | 75.7  | 72.8  | 65.1  | 58.4  | 47.5 |
|                                                                               |             | MCDB   | 49.8  | 54.5  | 63.7  | 71.2  | 75.0  | 80.6  | 84.4  | 84.4  | 80.1  | 70.3  | 62.8  | 52.4 |
|                                                                               | 10%         | WB     | 43.2  | 48.3  | 54.9  | 63.0  | 67.9  | 73.4  | 75.4  | 74.8  | 71.5  | 63.3  | 56.2  | 45.6 |
|                                                                               |             | MCDB   | 47.5  | 52.6  | 60.2  | 68.9  | 73.8  | 78.9  | 83.1  | 83.0  | 77.6  | 68.0  | 61.0  | 50.2 |
| Mean Daily<br>Temperature<br>Range                                            | 5% DB       | MDBR   | 9.1   | 9.9   | 12.1  | 13.3  | 13.3  | 12.2  | 13.5  | 14.2  | 12.6  | 11.0  | 10.9  | 10.2 |
|                                                                               |             | MCDBR  | 18.2  | 19.3  | 21.9  | 22.7  | 22.1  | 18.7  | 18.0  | 18.7  | 18.7  | 19.2  | 19.6  | 20.7 |
|                                                                               |             | MCWBR  | 10.0  | 10.7  | 10.3  | 9.4   | 7.8   | 6.2   | 4.8   | 4.7   | 6.0   | 7.1   | 8.9   | 10.8 |
|                                                                               | 5% WB       | MCDBR  | 15.5  | 14.9  | 17.9  | 17.0  | 14.1  | 13.0  | 14.7  | 16.0  | 14.6  | 13.4  | 14.2  | 16.4 |
|                                                                               |             | MCWBR  | 9.3   | 9.2   | 9.8   | 8.4   | 7.3   | 5.2   | 4.9   | 4.9   | 5.4   | 6.4   | 7.7   | 9.4  |
| Clear-Sky<br>Solar<br>Irradiance                                              | taub        | 0.679  | 0.705 | 0.742 | 0.710 | 0.619 | 0.593 | 0.547 | 0.599 | 0.631 | 0.691 | 0.695 | 0.676 |      |
|                                                                               | taud        | 1.452  | 1.437 | 1.416 | 1.516 | 1.738 | 1.840 | 1.981 | 1.820 | 1.706 | 1.523 | 1.480 | 1.463 |      |
|                                                                               | Ebn at Noon | 172    | 185   | 191   | 206   | 227   | 232   | 242   | 228   | 214   | 186   | 167   | 164   |      |
|                                                                               | Edh at Noon | 84     | 92    | 99    | 92    | 74    | 67    | 58    | 67    | 73    | 83    | 81    | 80    |      |
| All-Sky Solar<br>Radiation                                                    | RadAvg      | 469    | 567   | 829   | 1019  | 1097  | 1146  | 1473  | 1388  | 1064  | 728   | 604   | 470   |      |
|                                                                               | RadStd      | 100    | 133   | 89    | 155   | 209   | 184   | 163   | 149   | 143   | 129   | 125   | 104   |      |

## Historical Trends

|      | DBAvg                  | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------|------------------------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|      |                        | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|      |                        | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| (46) | Station Only           | N/A     | N/A    | N/A     | N/A   | -0.59 | -0.76       | +189  | +324  | N/A   |
| (47) | Regional (5 neighbors) | N/A     | N/A    | N/A     | +0.86 | N/A   | N/A         | N/A   | N/A   | +162  |

Nomenclature: See separate page